MODULE - 3.

NEAREST NEIGHBOR CLASSIFIERS:-

→ It is a supervised learning techniques

→ comes under classification problem.

→ The KNN algorithm stores available data and classifies new data based on the similarity.

→ KNN algorithm assumes the similarity between available data and new data,

→ Categories new data which is similar to available data.

→ It is also called as lazy learner algorithm, because it first stores the data and ~~pe~~ performs predictions when new data

Comes to model.

Why we need KNN algorithm?

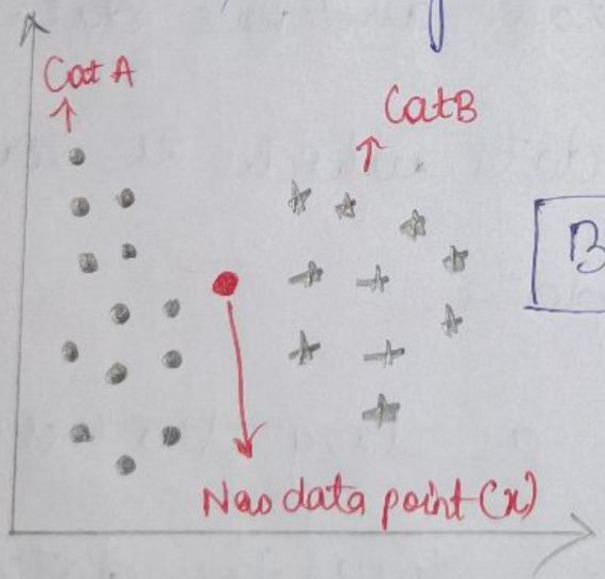
→ Suppose there are 2 categories

→ category A and category B.

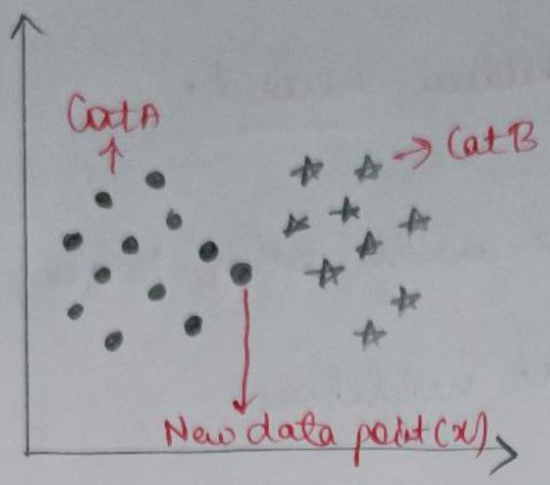
→ we have new data point (x)

→ this new data point belongs to which category?

→ To solve this type of problem we use KNN algorithm.

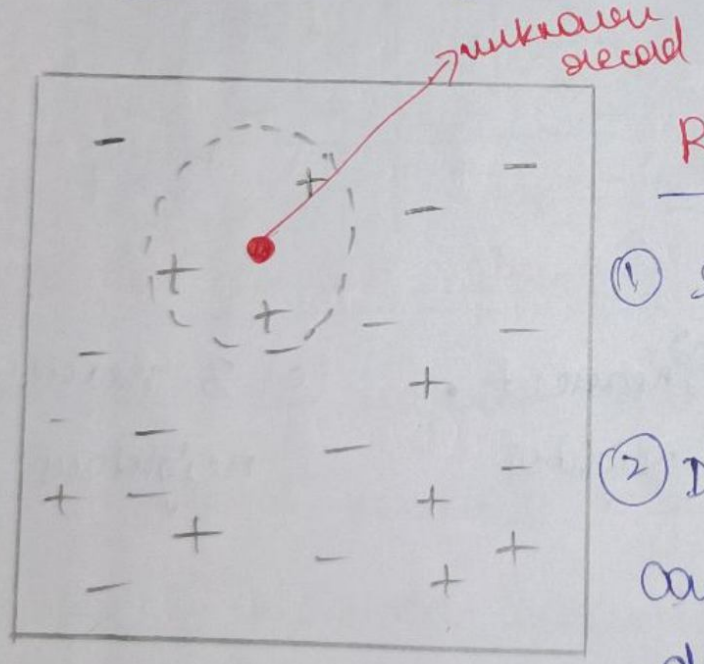


Before KNN



As we can see after applying KNN algorithm the new data point (x) belongs to Category A. based on the similarity with data.

Let's under KNN in depth:-

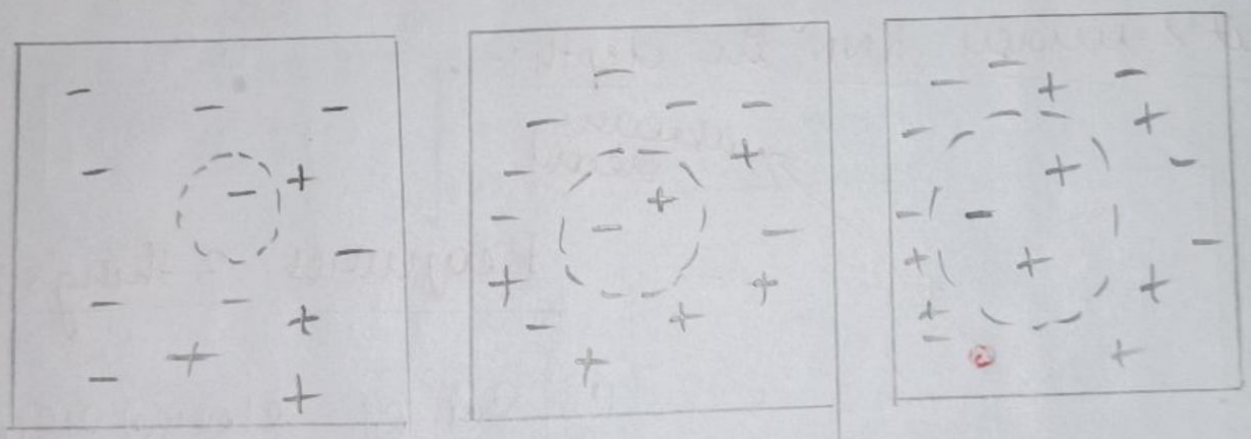


Requires 3 things:-

1. Set of stored records/ data
2. Distance metric to compute the distance b/w data points
3. The value of K, no. of nearest neighbour to retrieve

To classify an unknown record:

1. Compute distance with other training records
2. Identify K nearest neighbours
3. Use class labels of nearest neighbours to determine the class label of unknown record (Eg:- by taking majority vote)



(a) 1-nearest neighbor (b) 2-nearest neighbor (c) 3-nearest neighbour

→ K-nearest neighbours of a record x are data points that have the K smallest distance to x.

Distance Computation :-

→ Compute distance b/w 2 points.

→ Here we use Euclidean distance

Formula for euclidean distance :-

$$d(p,q) = \sqrt{\sum_{i} (p_i - q_i)^2}$$

Determine the class :-

→ take the majority vote of class labels among the K-nearest neighbors.

→ Weigh the vote according to distance

* weight factor, $w = 1/d^2$.

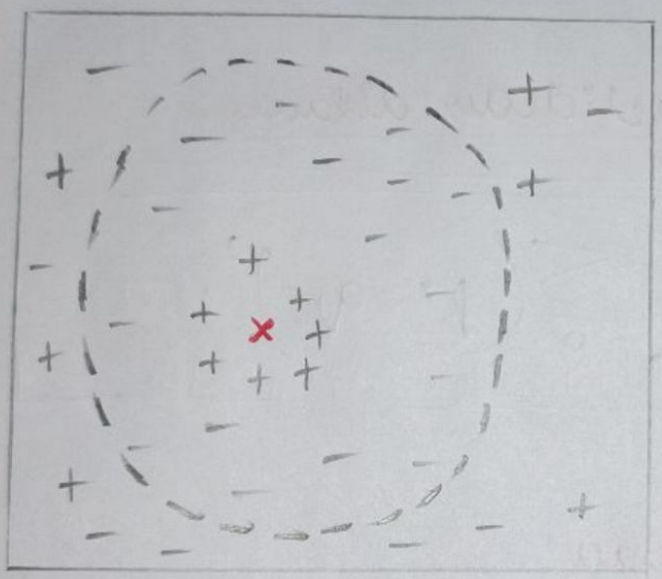
Choosing the value of K:-

→ If K is too small.

↳ leads to underfitting due to noise points in the training data.

→ If k is too large

↳ Neighbourhood may include points from other classes.



KNN algorithm steps:-

① Let k be the no. of nearest neighbor

② Let D be the set of training examples

③ for each test example $z = (x', y')$ do.

(4) Compute distance $d(x', x)$, the distance b/w z and every example $(x, y) \in D$.

(5) Select $D_z \subseteq D$, the set of k closest training examples to z.

(6) $$y' = \underset{v}{\operatorname{argmax}} \sum_{(x_i, y_i) \in D_z} I(v = y_i)$$

(7) End for.

Scaling Issues in KNN:-

→ Attributes may have to be scaled to prevent distance measures from being dominated by one of the attributes.

→ Eg:- (1) height of a person may vary from 1.5m to 1.8m

(2) weight of a person may vary from 90 lb to 300 lb

(3) income of a person may vary from \$10k to \$1M.

Solution:- Normalize the vectors to unit length.

KNN classifier are lazy learner:

→ It does not build models explicitly

→ Unlike eager learners such as decision tree and rule based system.

→ Classifying unknown records are relatively expensive.

Numerical Problem

① Select k-value

② Compute the Euclidean distance

③ Select the datapoints with minimum distance.

NOTE:- * Based on the K-value we are selected the data points will be classified into the class.

* If K = 3, then 3 nearest neighbour points are selected with minimum distance.

Problem:- Find ~~the~~ whether the given unknown data (student) is passed or failed using the below dataset.

Query/Condition:- $x = (Maths = 6,\ CS = 8)$.

When the student x gets 6 in maths & 8 in CS, we have to predict whether he/she will pass or fail.

Dataset:-

SI No	Maths	CS	Result
1	4	3	Fail
2	6	7	Pass
3	7	8	Pass
4	5	5	Fail
5	8	8	Pass

→ Let's take $K = 3$.

→ Euclidean distance formula:

Here we have take the difference between Observed value and the actual value.

$$\boxed{d(p,q) = \sqrt{\sum_{i} (p_i - q_i)^2}}$$

$$\boxed{d(p,q) = \sqrt{(P_{01} - q_{A1})^2 + (P_{02} - q_{02})^2}}$$

① $\sqrt{(6-4)^2 + (8-3)^2} = \sqrt{29} = 5.38$

② $\sqrt{(6-6)^2 + (8-7)^2} = 1$

③ $\sqrt{(6-7)^2 + (8-8)^2} = 1$

④ $\sqrt{(6-5)^2 + (8-5)^2} = \sqrt{10} = 3.16$

⑤ $\sqrt{(6-8)^2 + (8-8)^2} = 2.$

→ For the given query x, we will select 3 nearest neighbour. ⑥

→ For student (x), we ~~will be~~ have 3 nearest neighbor values.

→ we should take the data points which is having minimum.

→ The minimum distance between observed data and the actual data.

→ So therefore for the given problem we have 3 examples with minimum distance of Sl No 2, 3 and 5 (that is 1, 1, 2) and we have 0 Fail examples.

→ So we can conclude that the student x is also pass because all the neighbours belongs to pass class.

Advantages of KNN algorithm

1. It is simple to implement
2. It is robust to noisy training data
3. It can be more effective if the training data is large.

Disadvantages of KNN algorithm

1. always needs to determine the value of K, some times it may be complex
2. Computation cost high because of calculating the distance between the data points for all the training samples.